OpenEdge Bridge — Documentation

Version: 1.0
Date: July 23, 2026
Purpose: Bidirectional sync bridge between Progress OpenEdge and MySQL for Entrust Instant ID card printing.

Architecture Overview

```
┌──────────────────────────────────┐
│  Progress OpenEdge SQL Server    │  ← Source of truth (remote, provided by client)
└──────────────┬───────────────────┘
               │ ODBC
┌──────────────▼───────────────────┐
│  OpenEdge ODBC Driver            │  ← PROGRESS_DATADIRECT_ODBC_8.0_WIN_64_INSTALL.exe
│  (DataDirect 8.0, 64-bit)        │
└──────────────┬───────────────────┘
               │
┌──────────────▼───────────────────┐
│  C# Windows Service              │  ← OpenEdgeBridge.exe (this application)
│  (.NET 8, self-contained)        │
└──────────────┬───────────────────┘
               │ MySQL Connector
┌──────────────▼───────────────────┐
│  MySQL Server (XAMPP)            │  ← Local database, runs on the same PC
└──────────────┬───────────────────┘
               │ JDBC (mysql-connector-j-8.0.33.jar)
┌──────────────▼───────────────────┐
│  Entrust Instant ID              │  ← Card issuance system
└──────────────────────────────────┘
```

Data Flow

Phase 1: OpenEdge → MySQL (Primary Sync)

```
OpenEdge PUB.cardholder ──► C# Service ──► MySQL cardholders table
       (read)              (sync loop)         (upsert)
```

All cardholder fields synced:

- `ID` (OpenEdge primary key, stored as `openedge_id` in MySQL)
- `EMPLOYEE_ID`, `FIRST_NAME`, `LAST_NAME`, `DEPARTMENT`, `POSITION`
- `PRINT_COUNT`

Runs every N seconds (configurable in `appsettings.json`). Performs an **existence check** before each write — INSERT if new, UPDATE if already present. No duplicates.

Phase 2: MySQL → OpenEdge (Print Count Back-Sync)

```
MySQL cardholders ──► C# Service ──► OpenEdge PUB.cardholder
 (print_count)       (compare)         (UPDATE PRINT_COUNT)
```

Only `PRINT_COUNT` flows back. If MySQL has a higher count (updated by Entrust), the bridge increments the corresponding OpenEdge record to match.

Prerequisites

On the Target Windows PC

#	Component	Version / File	Notes
1	**Windows OS**	Windows 10/11 or Windows Server 2016+	64-bit required
2	**OpenEdge ODBC Driver**	`PROGRESS_DATADIRECT_ODBC_8.0_WIN_64_INSTALL.exe`	Install first; configure a System DSN
3	**XAMPP**	Latest (8.x)	Provides local MySQL server
4	**MySQL Connector/J**	`mysql-connector-j-8.0.33.jar`	For Entrust Instant ID to connect to MySQL
5	**OpenEdgeBridge**	Published `publish/` folder	Self-contained, no .NET runtime needed

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Step-by-Step Setup

1. Install OpenEdge ODBC Driver

Run the installer as Administrator:

```
PROGRESS_DATADIRECT_ODBC_8.0_WIN_64_INSTALL.exe
```

After installation, open **ODBC Data Sources (64-bit)** and create a **System DSN**:

1. Press `Win + R`, type `odbcad32.exe`, press Enter
2. Go to the **System DSN** tab
3. Click **Add...**
4. Select **Progress DataDirect OpenEdge SQL Server** (or similar)
5. Configure:
   - **Data Source Name:** e.g., `OpenEdgeInstantID`
   - **Host:** (provided by your OpenEdge team)
   - **Port:** (typically the OpenEdge SQL port)
   - **Database:** (provided by your OpenEdge team)
   - **User ID / Password:** (provided by your OpenEdge team)
6. Click **Test Connect** to verify, then **OK**

2. Install XAMPP

1. Download from https://www.apachefriends.org/
2. Run the installer with default settings

3. Open the **XAMPP Control Panel**
4. Click **Start** next to **MySQL**
5. Click **Admin** to open phpMyAdmin

3. Create the MySQL Database & Table

In phpMyAdmin (or any MySQL client), run:

```
CREATE DATABASE IF NOT EXISTS instantid_bridge
  CHARACTER SET utf8mb4
  COLLATE utf8mb4_unicode_ci;

USE instantid_bridge;

CREATE TABLE IF NOT EXISTS cardholders (
    openedge_id   INT          NOT NULL,
    employee_id   VARCHAR(50)  NOT NULL DEFAULT '',
    first_name    VARCHAR(100) NOT NULL DEFAULT '',
    last_name     VARCHAR(100) NOT NULL DEFAULT '',
    department    VARCHAR(100) NOT NULL DEFAULT '',
    position      VARCHAR(100) NOT NULL DEFAULT '',
    print_count   INT          NOT NULL DEFAULT 0,
    synced_at     DATETIME     NOT NULL DEFAULT CURRENT_TIMESTAMP,
    UNIQUE INDEX idx_openedge_id (openedge_id)
) ENGINE=InnoDB;
```

4. Configure Entrust Instant ID (MySQL Connector/J)

1. Copy `mysql-connector-j-8.0.33.jar` to Entrust's `lib/` folder
2. Configure Entrust to point to the local MySQL database:
   - **Host:** `localhost`
   - **Port:** `3306`
   - **Database:** `instantid_bridge`
   - **User:** `root` (or a dedicated user)
   - **Table:** `cardholders`

5. Deploy the Bridge Service

Copy the entire `publish/` folder to the target PC, e.g. `C:\OpenEdgeBridge\` .

Edit `appsettings.json`

```
{
  "ConnectionStrings": {
    "OpenEdge": "DSN=OpenEdgeInstantID;Uid=your_user;Pwd=your_password;",
    "MySql": "Server=localhost;Database=instantid_bridge;Uid=root;Pwd=your_mysql_password;"
  },
  "Sync": {
    "IntervalSeconds": 10
  }
}
```

Setting	Description
`OpenEdge`	ODBC DSN name, user, and password
`MySql`	MySQL server, database, user, and password
`Sync:IntervalSeconds`	How often the sync loop runs (default: 10s)

6. Install as a Windows Service

Open **PowerShell as Administrator** and run:

```
New-Service -Name "OpenEdgeBridge" `
            -BinaryPathName "C:\OpenEdgeBridge\OpenEdgeBridge.exe" `
            -DisplayName "OpenEdge Bridge Service" `
            -Description "Syncs cardholder data between OpenEdge and MySQL for Entrust
Instant ID" `
            -StartupType Automatic

Start-Service OpenEdgeBridge
```

Verify:

```
Get-Service OpenEdgeBridge
# Status should be: Running
```

Service Management Commands

```
Start-Service OpenEdgeBridge      # Start
Stop-Service OpenEdgeBridge       # Stop
Restart-Service OpenEdgeBridge    # Restart (after config changes)
Get-Service OpenEdgeBridge        # Check status

# Uninstall
Stop-Service OpenEdgeBridge
sc delete OpenEdgeBridge
```

Logging & Troubleshooting

Log Locations

Source	Location
File logs	`C:\OpenEdgeBridge\Logs\` (dated log files)
Windows Event Viewer	*Windows Logs* → *Application*, Source: `OpenEdgeBridge`

Common Issues

Problem	Likely Cause	Fix
Service starts then stops immediately	Wrong DSN name or credentials	Test the ODBC DSN in ODBC Administrator with **Test Connect**
"MySQL connection string not found"	`appsettings.json` not in same folder as `.exe`	Ensure `appsettings.json` is in `C:\OpenEdgeBridge\`
Duplicate rows in MySQL	Missing UNIQUE index	Run the `ALTER TABLE ... ADD UNIQUE INDEX` SQL above
OpenEdge ODBC driver not found	Wrong bitness (32 vs 64)	Use `odbcad32.exe` (64-bit), not the 32-bit one in SysWOW64
MySQL not accessible from Entrust	MySQL not running or wrong port	Check XAMPP Control Panel; MySQL must be **Started**

Sync Interval Tuning

The default sync interval is **10 seconds**. Adjust in `appsettings.json`:

```
"Sync": {
  "IntervalSeconds": 30
}
```

- **Lower** (e.g., 5s) = faster sync, more database load
- **Higher** (e.g., 60s) = slower sync, less load
- Recommended: 10–30 seconds for production

Port Summary

Component	Port	Direction
OpenEdge SQL Server	Provided by OpenEdge team	Outbound
MySQL (XAMPP)	3306	Local
Entrust → MySQL	3306	Local via JDBC

No inbound ports are opened by the bridge itself.

Upgrade Procedure

1. Stop the service: `Stop-Service OpenEdgeBridge`
2. Replace all files in `C:\OpenEdgeBridge\` with the new `publish/` output
3. Preserve your `appsettings.json` (do not overwrite with default)
4. Start the service: `Start-Service OpenEdgeBridge`

Checklist for Target PC

- [ ] Windows 10/11 or Server 2016+ (64-bit)
- [ ] OpenEdge ODBC driver installed and DSN configured
- [ ] XAMPP installed and MySQL running
- [ ] `instantid_bridge` database and `cardholders` table created
- [ ] `mysql-connector-j-8.0.33.jar` placed in Entrust's lib folder
- [ ] Bridge `publish/` folder copied to `C:\OpenEdgeBridge\`
- [ ] `appsettings.json` updated with correct connection strings
- [ ] Windows Service installed and started
- [ ] Logs checked for errors